

C#.NET @ Sunbeam Infotech

Trainer: Nilesh Ghule

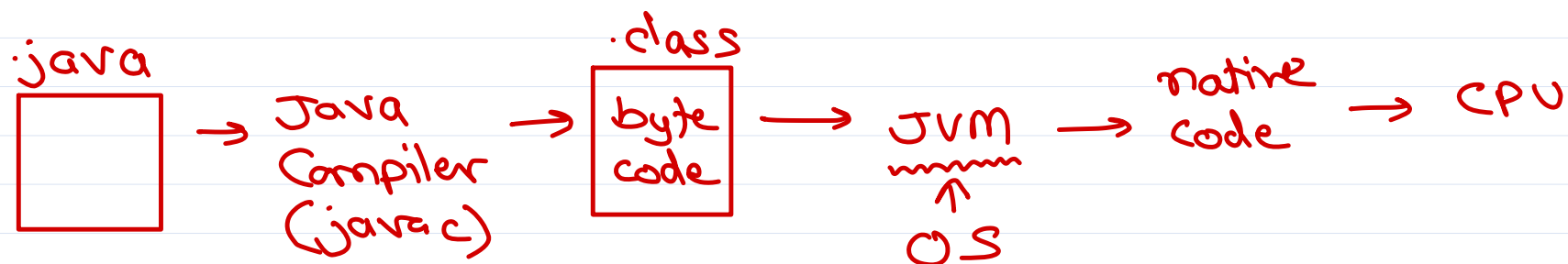


Agenda

- ① Java code compilation vs .Net lang code compilation
- ② C# code vs VB.Net code compilation
- ③ .Net framework & versions.
- ④ CTS, CLS, CLR (JIT, GC, ...)
- ⑤ Installations - on client machine vs on dev machine
- ⑥ Hello world appln (using Visual Studio)
 ↗ Console app
 ↘ Class Lib
- ⑦ .Net assembly (structure, exe execution) → ildasm, SharpLab.io, ...
- ⑧ Class libs & Namespaces
- ⑨ C# classes (fields, methods, properties, constructor, destructor, ...)

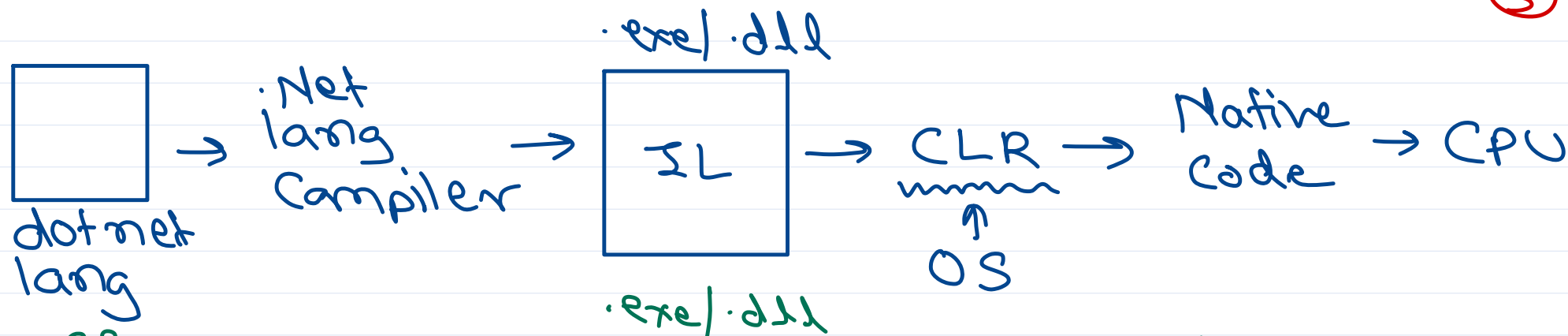


.Net - Compilation



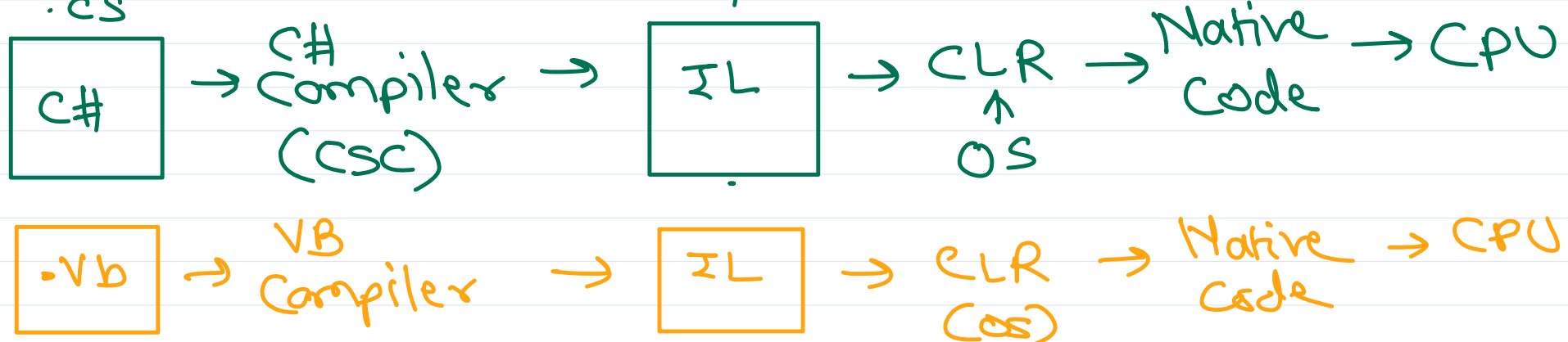
JVM based lang

- ① Java
- ② Kotlin
- ③ Scala

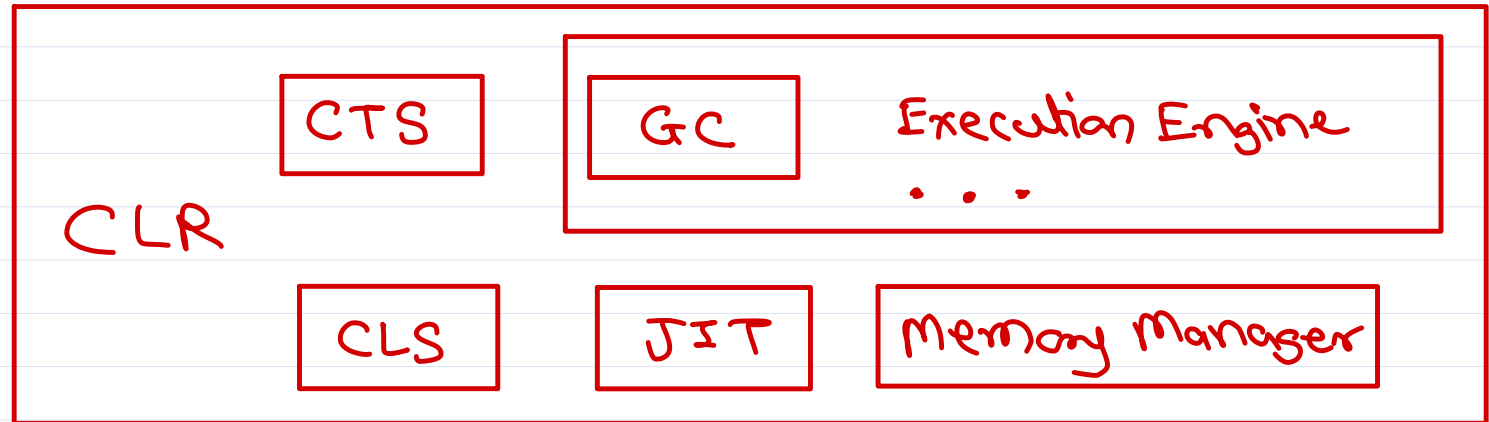
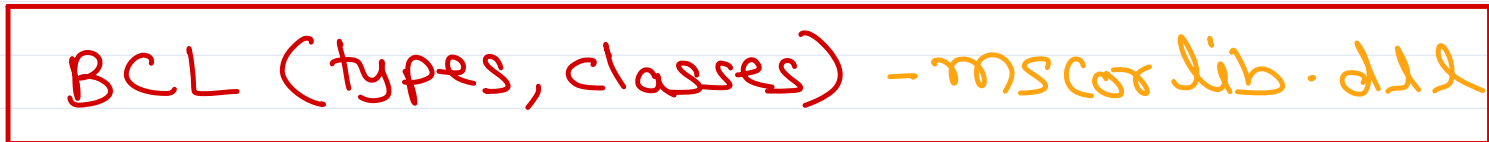
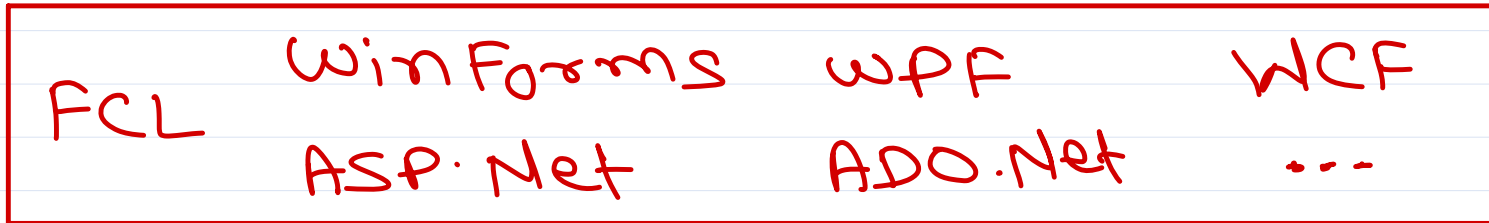
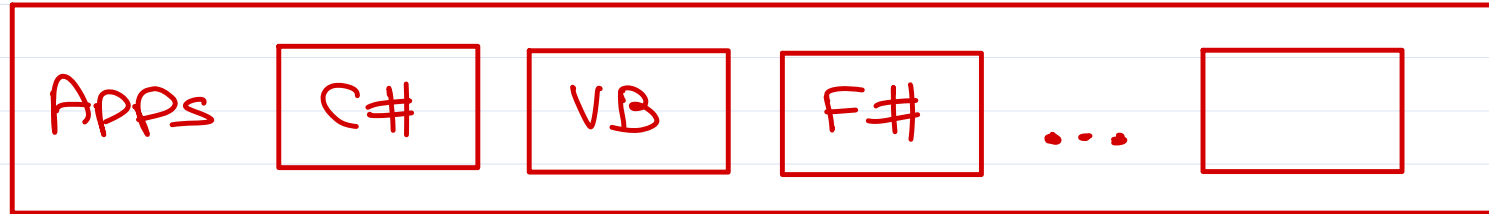


.Net lang

- ① C# (*)
- ② F#
- ③ VB.Net
- ④ J# x
- ⑤ Python
- ⑥ ...



.Net Framework



client machine Setup

- Net framework
 - ↳ CLR + BCL + FCL
- ## Dev machine Setup

.Net framework SDK

- Net framework
 - ↳ CLR + BCL + FCL
- + Dev tools
 - ↳ Lang Compilers + Debugger
 - + tools (ildasm, CAs, ...)
- + IDE (RAD tools) + ...

Visual Studio 2026  Community, Professional, Enterprise



Create a new project

Recent project templates

-  Console App
-  Windows Forms App
-  Console App (.NET Framework)
-  Class Library (.NET Framework)
-  Blank Solution

C#

C#

C#

C#

Search for templates (Alt+S)



[Clear all](#)

C#

Windows

Console



Console App

A project for creating a command-line application that can run on .NET on Windows, Linux and macOS

C#

Linux

macOS

Windows

Console



Console App (.NET Framework)

A project for creating a command-line application

C#

Windows

Console

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Next

Configure your new project

Console App (.NET Framework)

C#

Windows

Console

Project name

HelloWorld

Location

D:\Nilesh\Feb25\DAC\dotnet\day01\

Solution name

HelloWorld

☒ Place solution and project in the same directory

Framework

.NET Framework 4.8

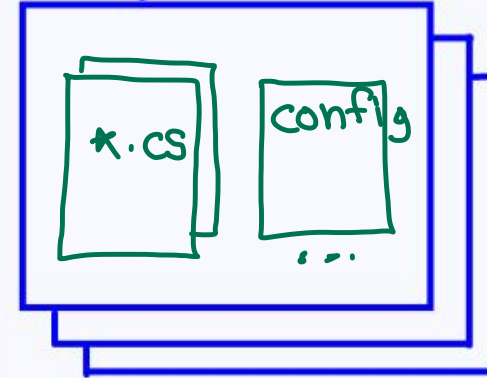
Project will be created in "D:\Nilesh\Feb25\DAC\dotnet\day01\HelloWorld"

Solution Info -> .sln file

Project Info -> .csproj file

For very simple project
1 soln -> only 1 project
Solution

Project



Solution = Group of
related projects
e.g. Business logic,
Win App, Web App.

Back

Create



Search



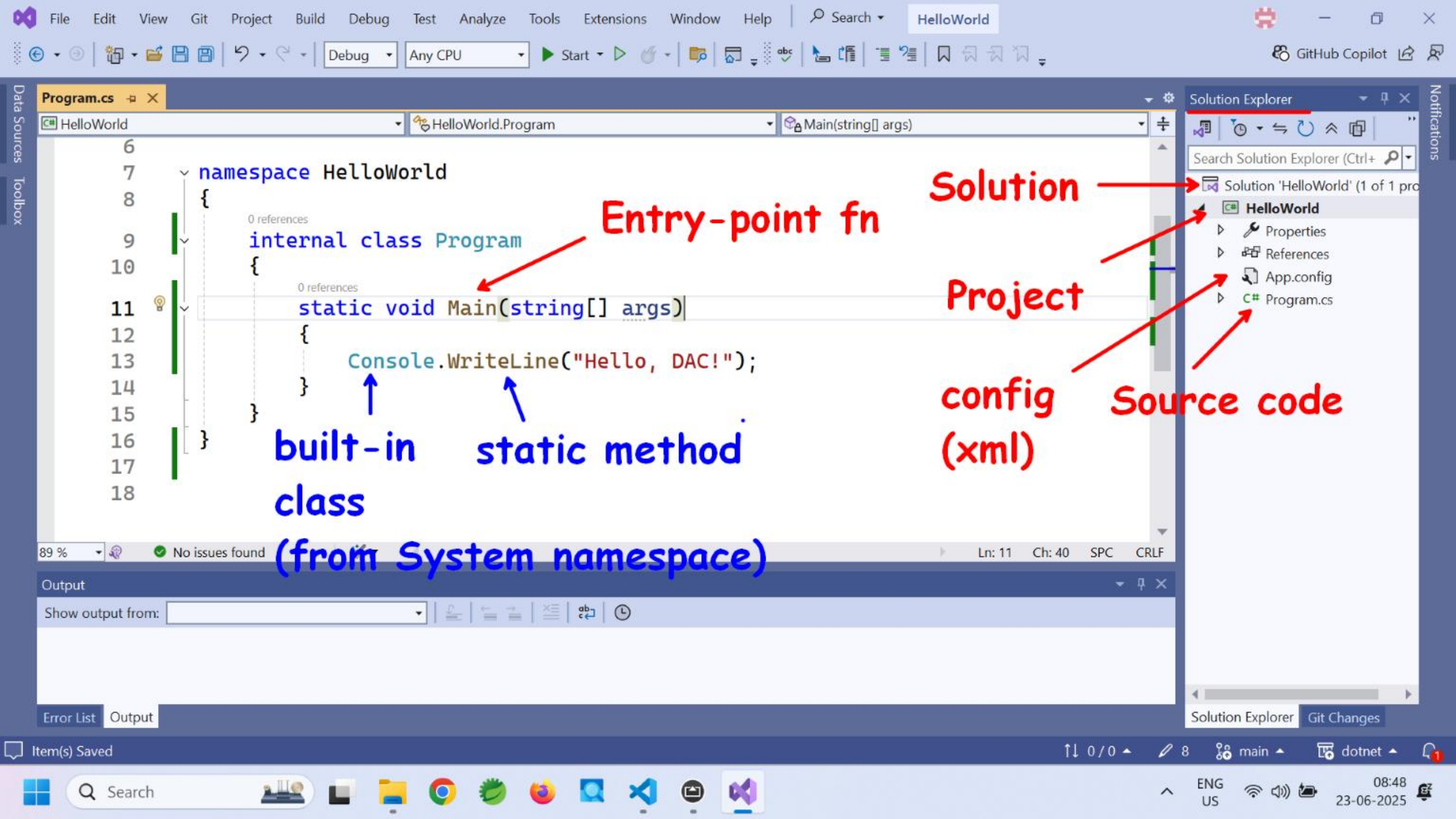
ENG
US



08:40

23-06-2025





Program.cs

HelloWorld

HelloWorld.Program

Main(string[] args)

Solution Explorer

Search Solution Explorer (Ctrl+)

Solution 'HelloWorld' (1 of 1 projects)

HelloWorld

Properties

References

App.config

Program.cs

Entry-point fn

Solution

Project

config
(xml)

Source code

built-in
class

static method

(from System namespace)

89 % No issues found

Ln: 11 Ch: 40 SPC CRLF

Output

Show output from:

Error List Output

Solution Explorer Git Changes

Item(s) Saved

0/0

8

main

dotnet

1



Search

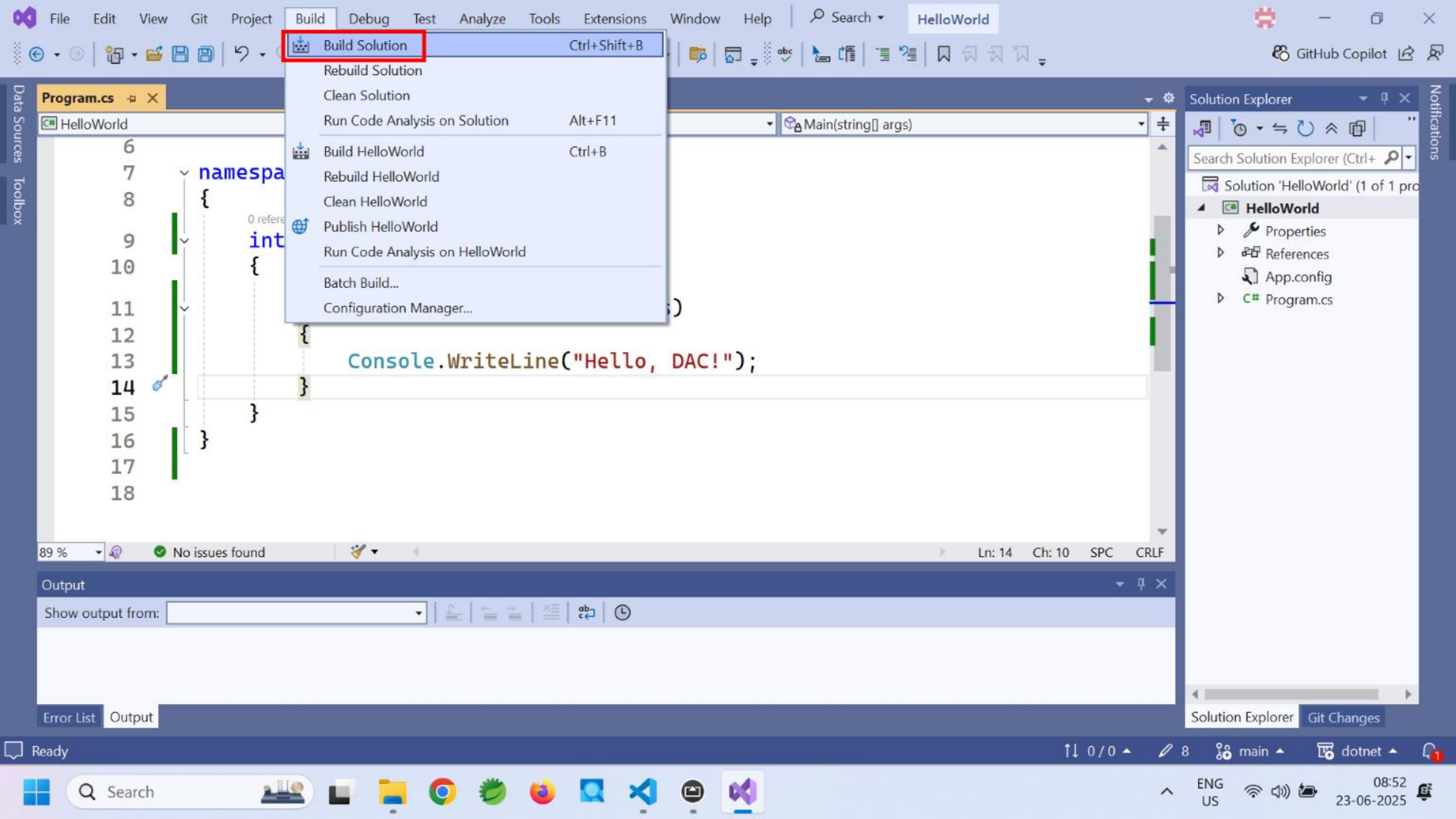


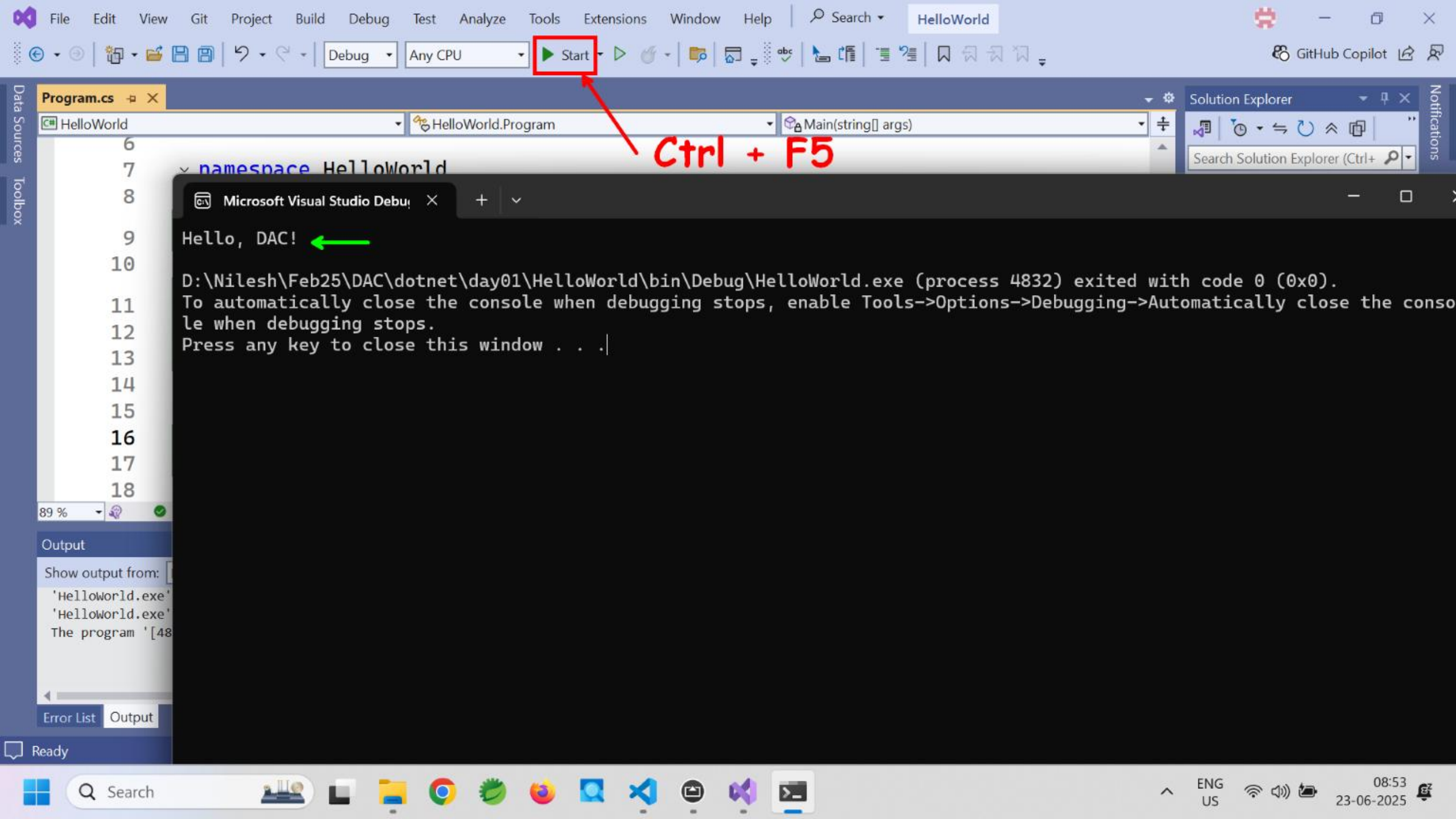
ENG
US

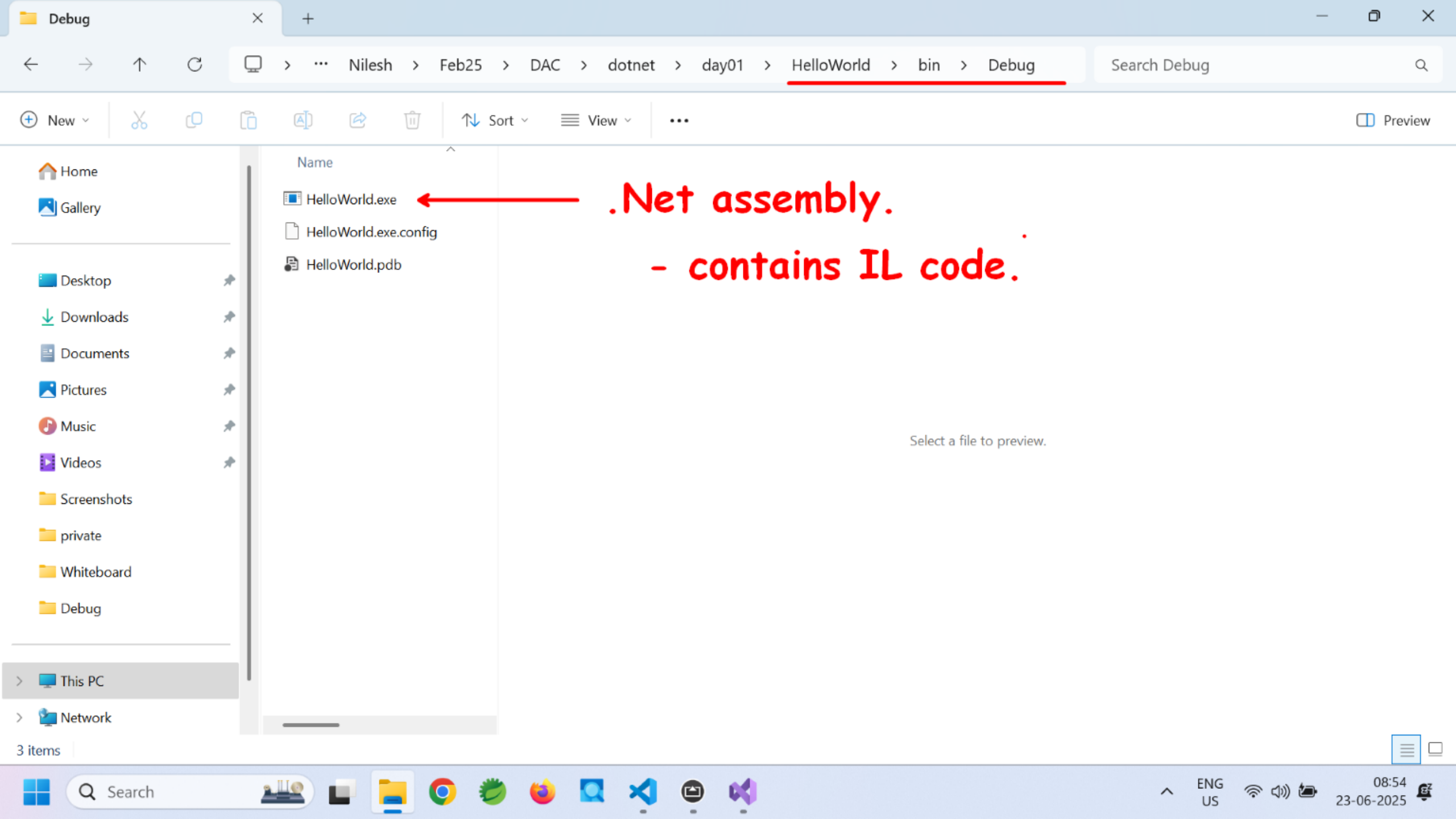


08:48
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.Net assembly.
- contains IL code.

FileEditViewGitProjectBuildDebugTestAnalyzeToolsExtensionsWindowHelp

SearchHelloWorld

GitHub Copilot

DebugAny CPUStart

Program.cs

HelloWorldHelloWorld.ProgramMain(string[] args)

12{

13

14

15

16

17

18

19}

20

21

22

CTSC# Language code

int num1 = 22, num2 = 7, result;

result = num1 + num2;

Console.WriteLine("Result: " + result);

.exe

IL

VB.NET code

Dim num1 As Integer

Dim num2 As Integer

Dim num3 As Integer

num1 = 22

num2 = 7

num3 = num1 + num2

Console.WriteLine(num3)

.exe

IL

CLR

Int32

Int64

Int16

Int8

String

Double

...

89 %No issues found

Ln: 10Ch: 10SPC

Output

Show output from: Build

Error List

Output

Solution Explorer

Search Solution Explorer (Ctrl+)

Solution 'HelloWorld' (1 of 1 projects)

HelloWorld

Properties

References

App.config

C# Program.cs

Item(s) Saved

0/0

9

main

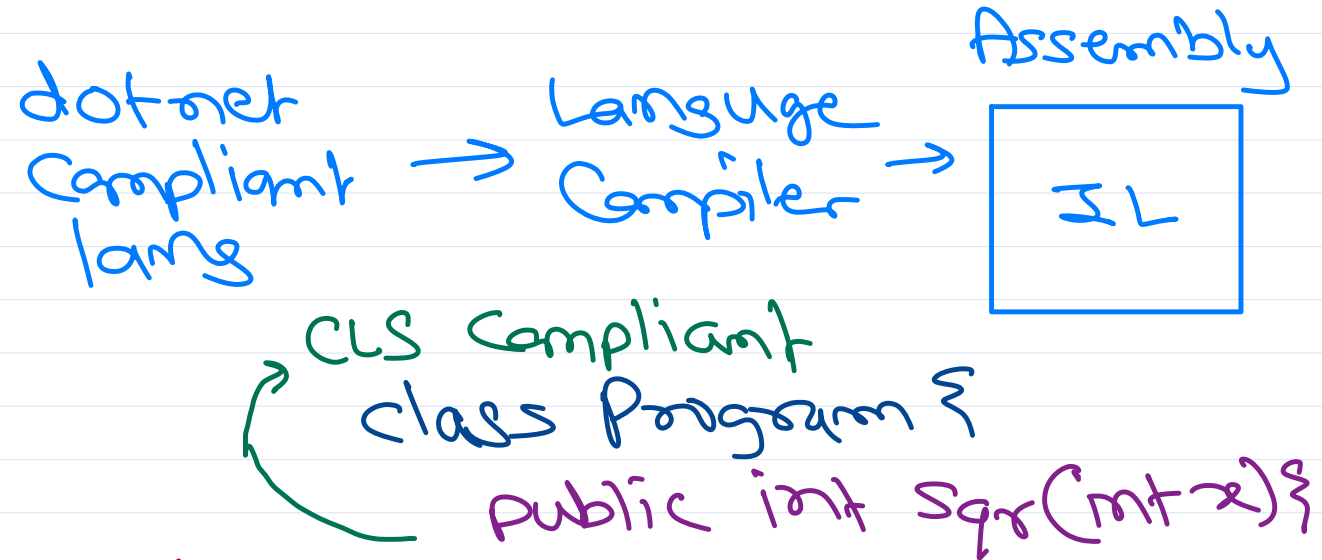
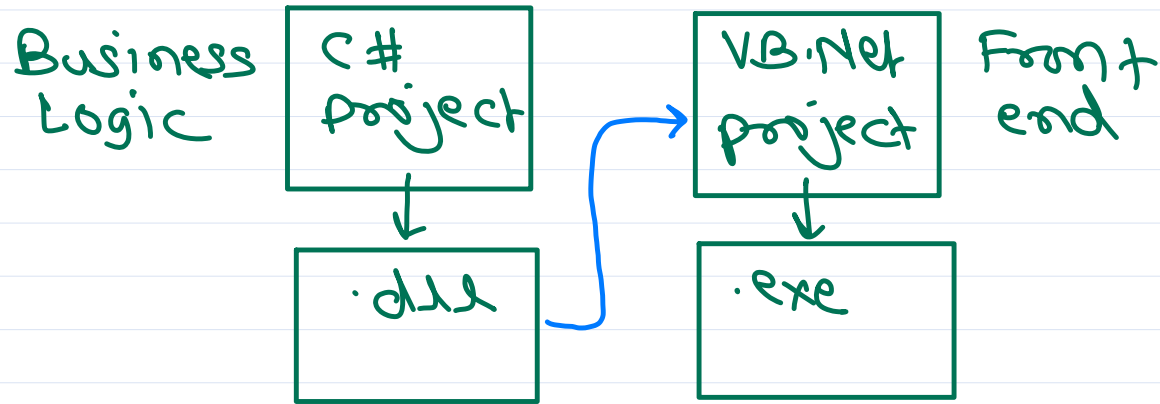
dotnet

09:37

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CLR — CTS & CLS

* Language interoperability



- * Languages must follow certain rules.
 - .Net Compliant lang → lang Compiler
 - lang rules → CLS
 - objects allocated using new
 - only signed types
 - stack based execution
 - ...

```
public int Sqr(int x) {  
return x * x;  
}
```

CLS Non-compliant



CLR — CTS & CLS

Common Type System → Types 'in IL → understood by CLR

CTS types

→ Value Types → represent value

Int32 a=123; a 123

① Int16 ⑥ Decimal

② Int32 ⑦ Boolean

③ Int64 ⑧ Char

④ Single

⑤ Double

* User defined

* Typically on stack (locals)

* If part/field in some obj
→ on Heap. CTS also have

Structure unsigned types
Enum e.g. UInt32, ...

→ Reference Types → represent

① Class (e.g. String, ...)

② Interface

③ Array

④ delegate

⑤ object

ref / addr.

String s = "DAC";

s
1000

"DAC"

1000

String obj



🔍 developer PowerShell for VS 2022

- ← All Apps Documents Web Settings Folders Photo 0 🔗 G ... 🌈

Best match

 Developer PowerShell for VS 2022
App

Apps

 Visual Studio 2022 >

 Developer Command Prompt for VS 2022 >

 Windows Software Development Kit >

Search the web

🔍 deve - See more search results >

Settings (7+)

Documents (1+)



Developer PowerShell for VS 2022
App

- 📄 Open
-  Run as Administrator
-  Run ISE as Administrator
-  Windows PowerShell ISE

Select a file to preview.

Debug

Search Debug

File View Help

HelloWorld.exe - IL DASM

assembly metadata

class members - IL code.

class (metadata + IL)

path of assembly.

On dev cmd prompt/power shell, the PATH and Env for all .NET tools (e.g. ildasm, csc, ...) is already set.

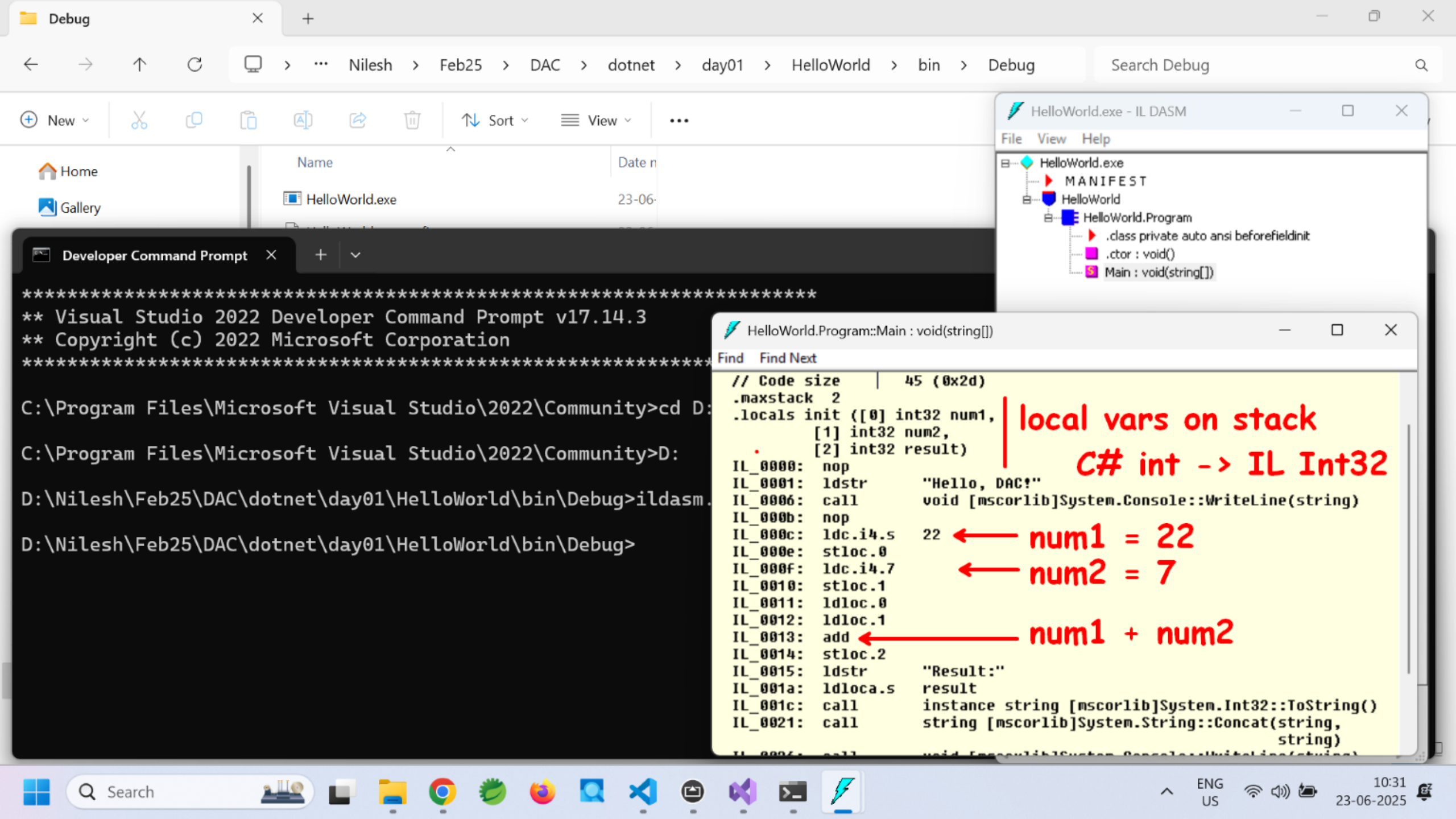
```
*****
** Visual Studio 2022 Developer Command Prompt v17.14.3
** Copyright (c) 2022 Microsoft Corporation
*****

C:\Program Files\Microsoft Visual Studio\2022\Community>cd D:\Nilesh\Feb25\DAC\dotnet\
C:\Program Files\Microsoft Visual Studio\2022\Community>D:
D:\Nilesh\Feb25\DAC\dotnet\day01\HelloWorld\bin\Debug>ildasm.exe HelloWorld.exe
D:\Nilesh\Feb25\DAC\dotnet\day01\HelloWorld\bin\Debug>
```

assembly HelloWorld

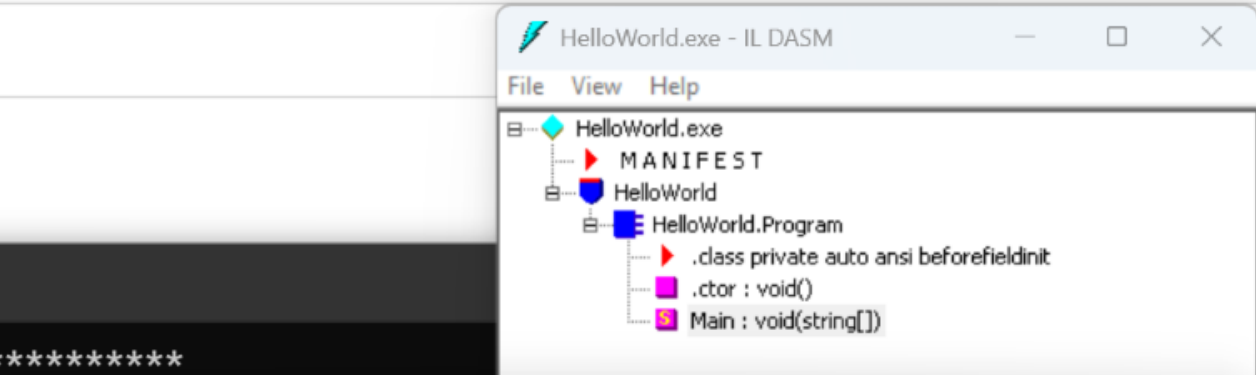
```
{
.ver 1:0:0:0
```

10:25 23-06-2025



```
*****
** Visual Studio 2022 Developer Command Prompt v17.14.3
** Copyright (c) 2022 Microsoft Corporation
*****

C:\Program Files\Microsoft Visual Studio\2022\Community>cd D:
C:\Program Files\Microsoft Visual Studio\2022\Community>D:
D:\Nilesh\Feb25\dac\dotnet\day01\HelloWorld\bin\Debug>ildasm
D:\Nilesh\Feb25\dac\dotnet\day01\HelloWorld\bin\Debug>
```

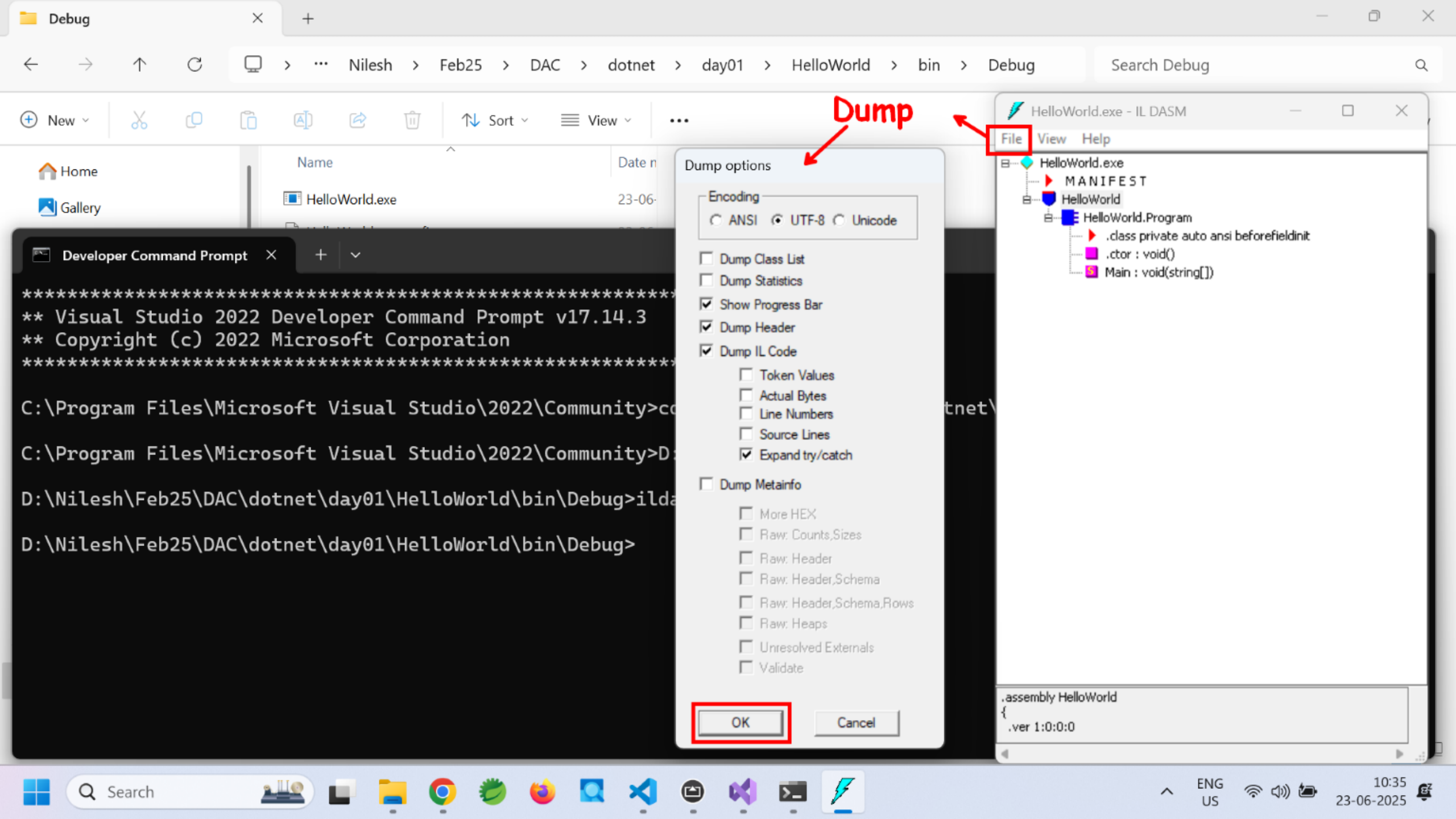


HelloWorld.Program::Main : void(string[])

Find Find Next

IL	Code	Size	Comment
IL_0000	nop		
IL_0001	ldstr		"Hello, DAC!"
IL_0006	call		void [mscorlib]System.Console::WriteLine(string)
IL_000b	nop		
IL_000c	ldc.i4.5	22	← num1 = 22
IL_000e	stloc.0		
IL_000f	ldc.i4.7		← num2 = 7
IL_0010	stloc.1		
IL_0011	ldloc.0		
IL_0012	ldloc.1		
IL_0013	add		← num1 + num2
IL_0014	stloc.2		
IL_0015	ldstr		"Result:"
IL_001a	ldloc.s		result
IL_001c	call		instance string [mscorlib]System.Int32::ToString()
IL_0021	call		string [mscorlib]System.String::Concat(string, string)

local vars on stack
C# int -> IL Int32



* JIT

- ✓ IL code → JIT → Native code
- ✓ Does conversion Method by Method.
- ✓ Frequently called methods - native code - Cached.
- ✓ JIT types
 - ① Standard JIT (Part of CLR) - all devices.
 - ② Economy JIT - on small devices (phones, embedded).
 - .Net Compact framework - Outdated.
 - ③ Pre-JIT (AOT - Ahead of Time):
 - all IL code → pre-JIT → Native code
 - ngen.exe ↓
 - Before Execution ↘



* GC → auto deallocate
unreferenced objects

- Mark-Sweep-Compact algorithm.

* Execution Engine

- ↳ GC
- ↳ Security

* Memory Manager
↳ allocates memory for

- class loading
- Stack
- Cache
- ...

.Net Assemblies

File Extensions

① .exe

- executable file
- OS creates process when end user starts execution.

② .dll

- lib of classes (reuse)
- dependent executable
- loaded in calling process memory, when its classes accessed in that process.

Assembly Structure

PE header
Manifest
Metadata
IL code
Resources

→ CLR info, referenced assms info, versions, ...

→ a.k.a. assembly metadata

→ type metadata - type name, fields, methods, ctors, ...

→ images, config files, ...



Create a new project

Recent project templates

 Console App (.NET Framework)

C#

 Console App

C#

 Windows Forms App

C#

 Class Library (.NET Framework)

C#

 Blank Solution

Solution

[Clear all](#)

C#

Windows

Console

No exact matches found

Other results based on your search



Blank Solution

Create an empty solution containing no projects

Other

Not finding what you're looking for?

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Next



Search



ENG
US



11:37
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Configure your new project

Blank Solution

Other

Solution name

MathSoln

Location

D:\Nilesh\Feb25\DAC\dotnet\day01

Solution

Create new solution

Solution will be created in "D:\Nilesh\Feb25\DAC\dotnet\day01\MathSoln\"

Back

Create



Search



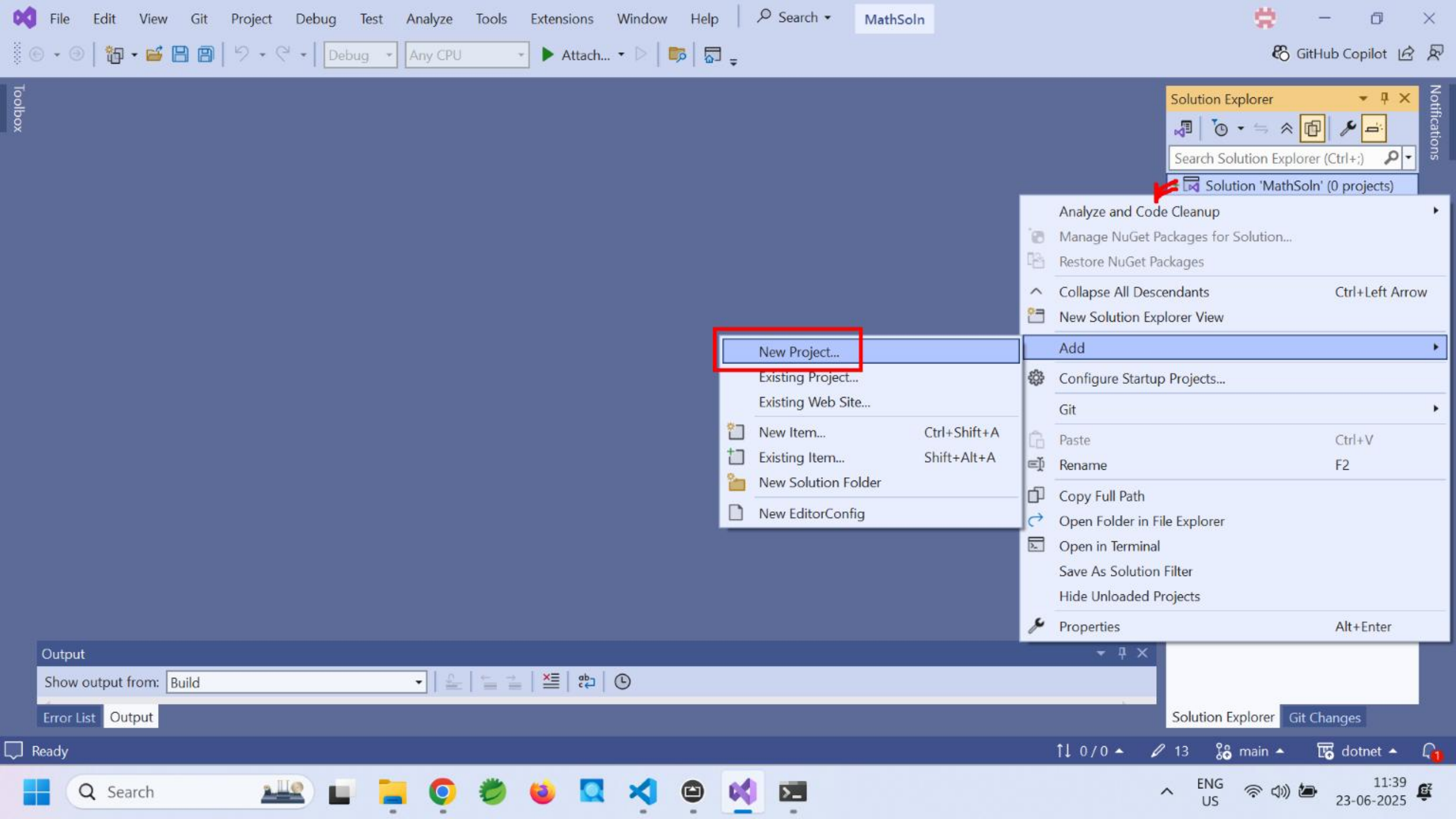
ENG
US



11:38

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New Project...

Add a new project

Recent project templates

 Console App (.NET Framework)

C#

 Console App

C#

 Windows Forms App

C#

 Class Library (.NET Framework)

C#

Search for templates (Alt+S)



Clear all

C#

Windows

Library

C#

Linux

macOS

Windows

Library

Web



WPF Class Library

A project for creating a class library that targets a .NET WPF Application

C#

Windows

Desktop

Library



WPF Custom Control Library

A project for creating a custom control library for .NET WPF Applications

C#

Windows

Desktop

Library



WPF User Control Library

A project for creating a user control library for .NET WPF Applications

C#

Windows

Desktop

Library



Class Library (.NET Framework)

A project for creating a C# class library (.dll)

C#

Windows

Library



WPF Custom Control Library (.NET Framework)

Windows Presentation Foundation custom control library

C#

XAML

Windows

Desktop

Library

Next



Search



ENG
US



11:40
23-06-2025



Configure your new project

Class Library (.NET Framework)

C#

Windows

Library

Project name

MathLib

Location

D:\Nilesh\Feb25\DAC\dotnet\day01\MathSoln

Framework

.NET Framework 4.8

Project will be created in "D:\Nilesh\Feb25\DAC\dotnet\day01\MathSoln\MathLib\"

Back

Create



Search

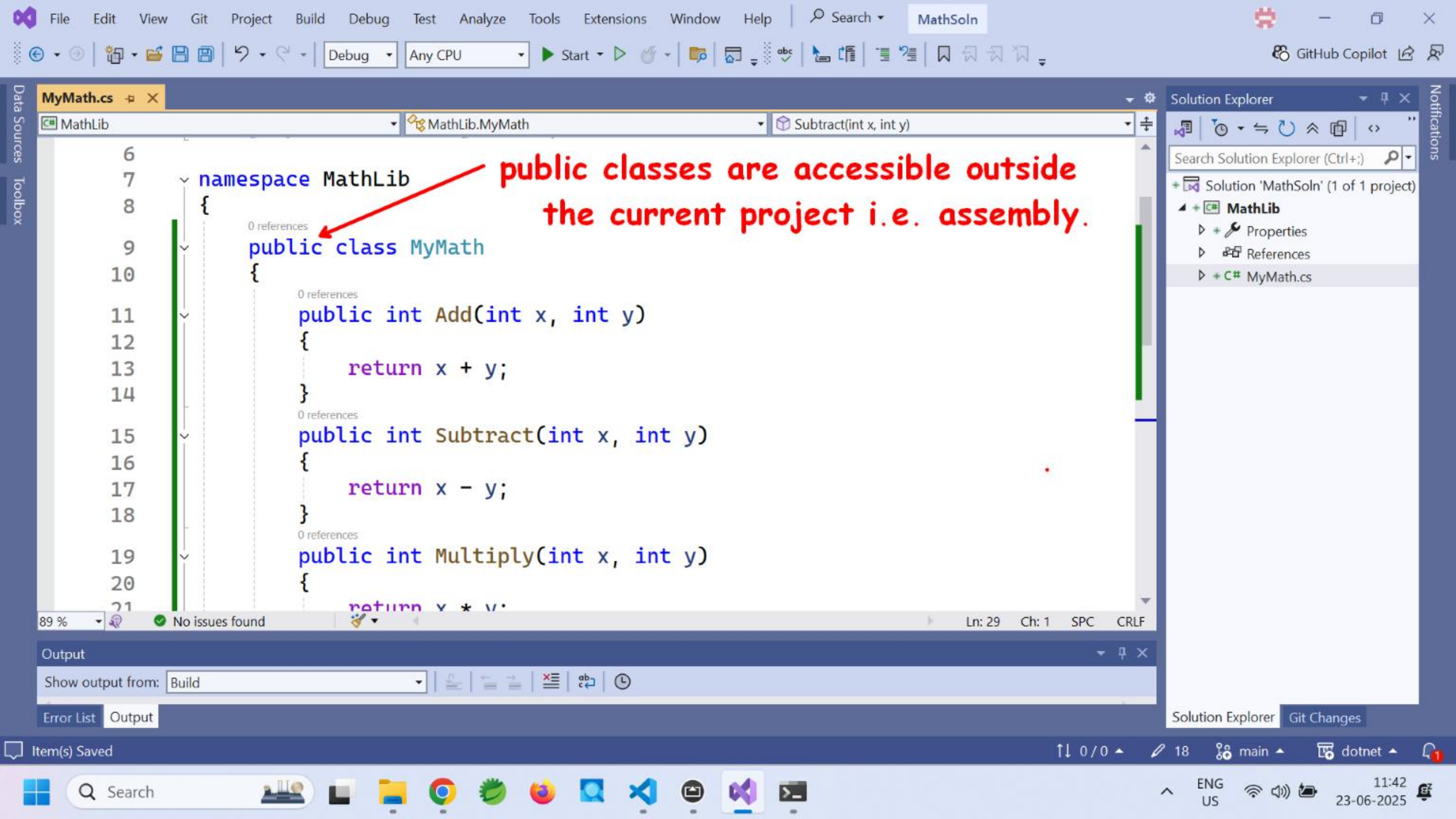


ENG
US



11:40
23-06-2025





public classes are accessible outside
the current project i.e. assembly.

```
namespace MathLib
```

```
{  
    public class MyMath
```

```
{  
    public int Add(int x, int y)
```

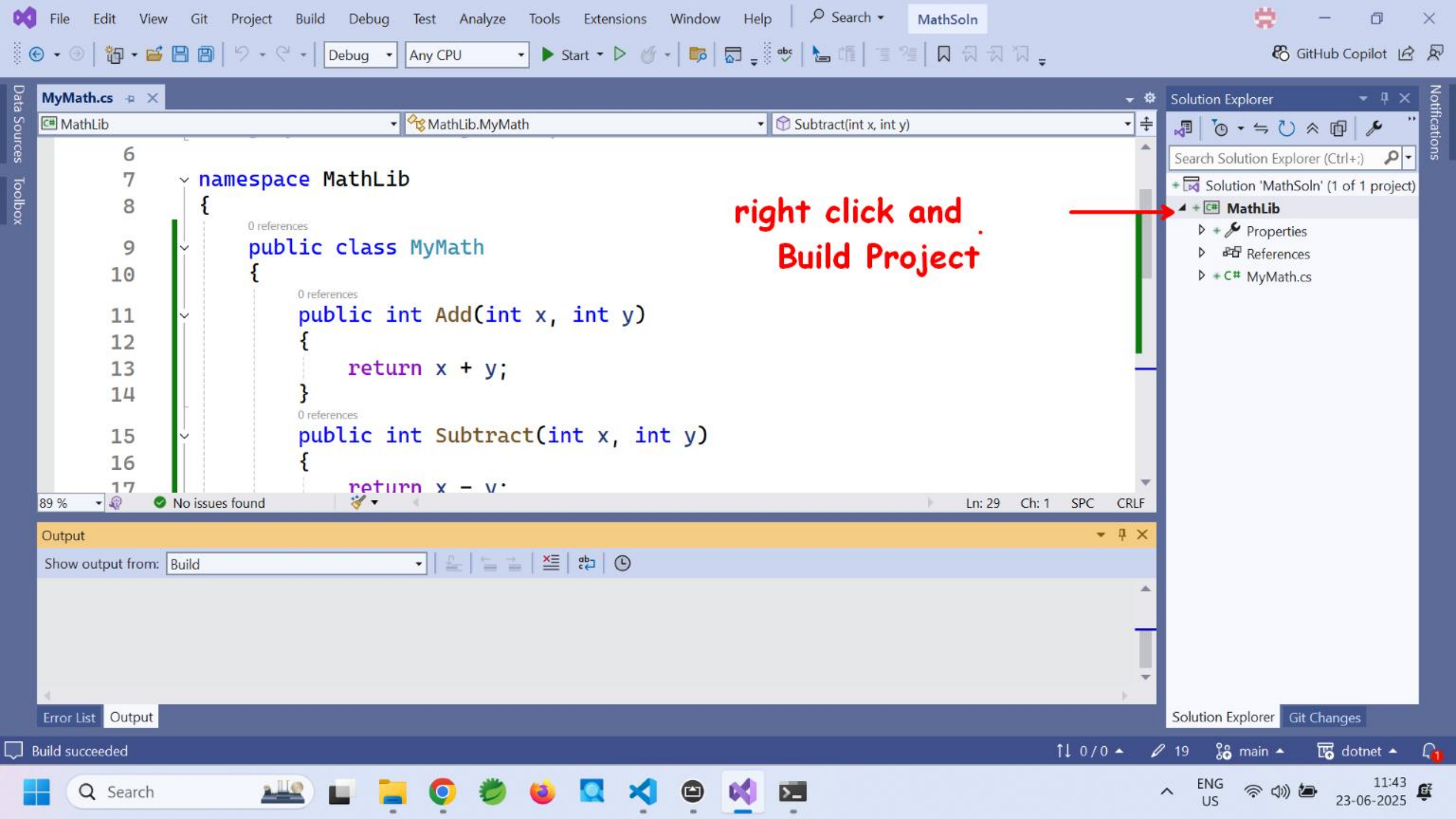
```
{  
        return x + y;
```

```
}  
    public int Subtract(int x, int y)
```

```
{  
        return x - y;
```

```
}  
    public int Multiply(int x, int y)
```

```
{  
        return x * y;
```

MyMath.cs

MathLib

MathLib.MyMath

Subtract(int x, int y)

```
6
7 namespace MathLib
8 {
9     0 references
10     public class MyMath
11     {
12         0 references
13         public int Add(int x, int y)
14         {
15             return x + y;
16         }
17         0 references
18         public int Subtract(int x, int y)
19         {
20             return x - y;
21         }
22     }
23 }
```

right click and
Build Project

Solution Explorer

Search Solution Explorer (Ctrl+;) 🔍

+ Solution 'MathSoln' (1 of 1 project)

- + C# MathLib
 - + Properties
 - + References
 - + C# MyMath.cs

Output

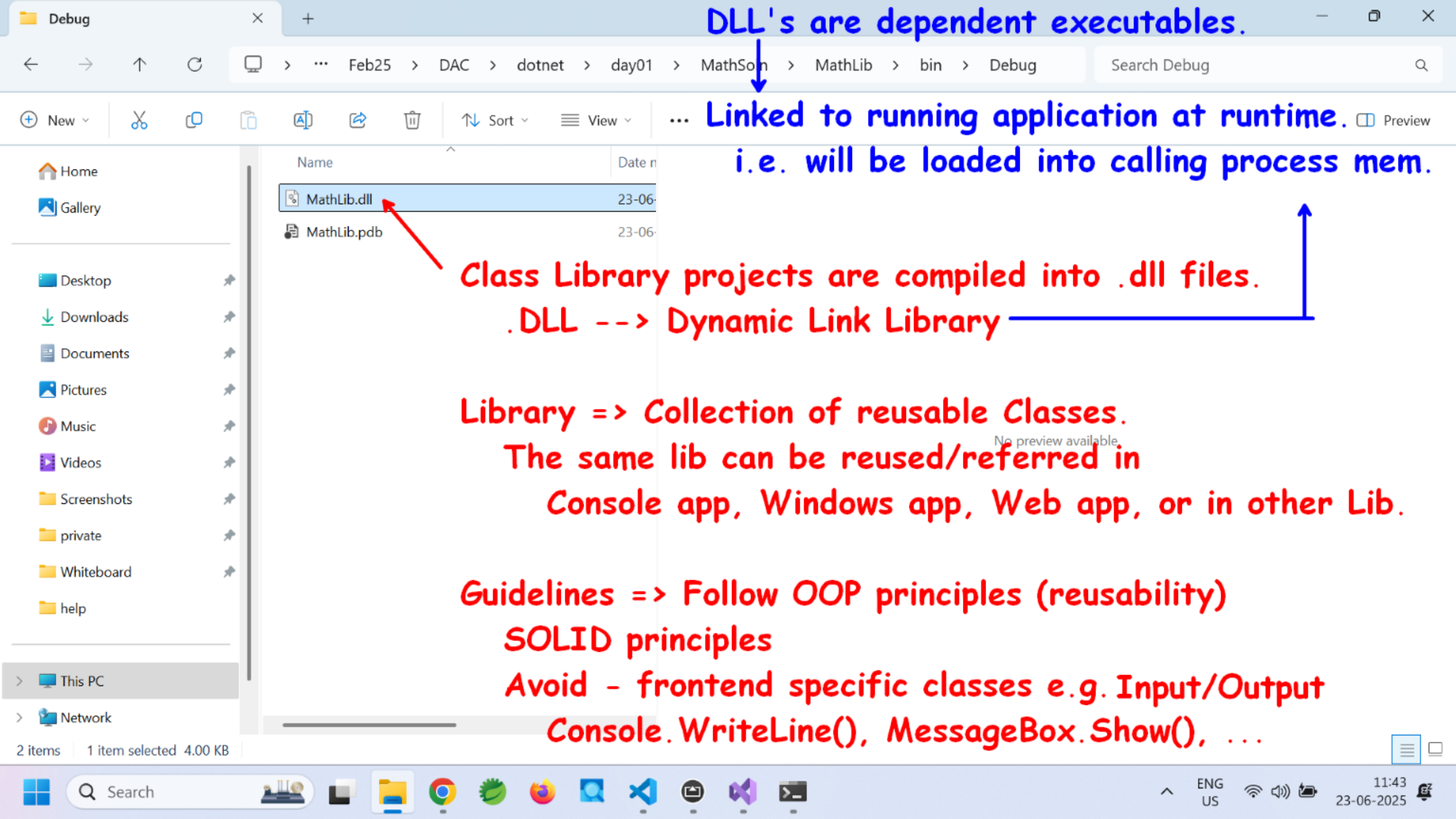
Show output from: Build

Error List Output

Solution Explorer Git Changes

Build succeeded

0/0 19 main dotnet



DLL's are dependent executables.

Linked to running application at runtime.

i.e. will be loaded into calling process mem.

Class Library projects are compiled into .dll files.

.DLL --> Dynamic Link Library

Library => Collection of reusable Classes.

The same lib can be reused/referred in

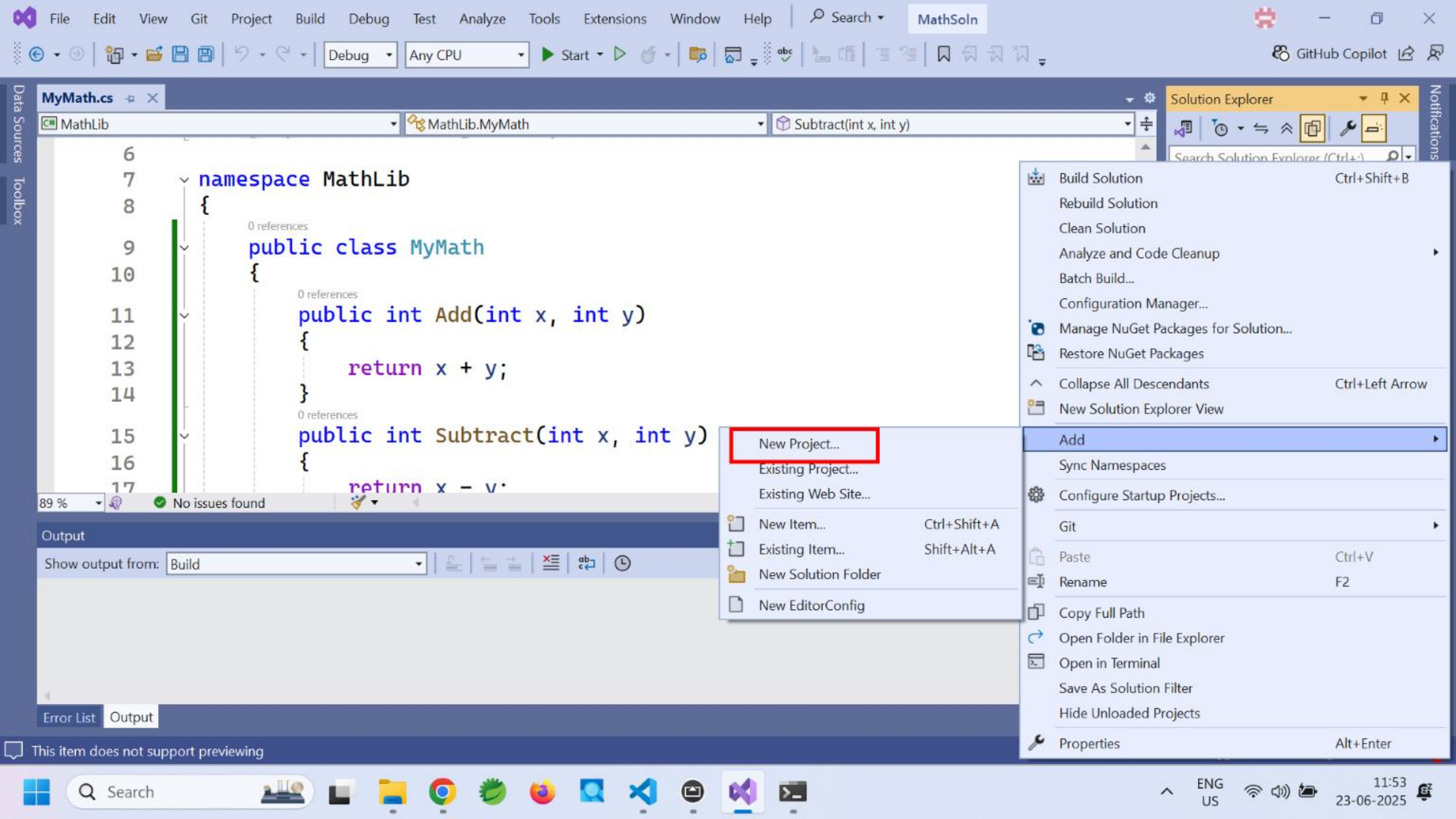
Console app, Windows app, Web app, or in other Lib.

Guidelines => Follow OOP principles (reusability)

SOLID principles

Avoid - frontend specific classes e.g. Input/Output

Console.WriteLine(), MessageBox.Show(), ...



Add a new project

Recent project templates

-  Class Library (.NET Framework)
-  Console App (.NET Framework)
-  Console App
-  Windows Forms App

Search for templates (Alt+S)



Clear all

C#

Windows

Console



Console App

A project for creating a command-line application that can run on .NET on Windows, Linux and macOS

C#

Linux

macOS

Windows

Console



Console App (.NET Framework)

A project for creating a command-line application

C#

Windows

Console

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Next

Configure your new project

Console App (.NET Framework)

C#

Windows

Console

Project name

MathClient

Location

D:\Nilesh\Feb25\DAC\dotnet\day01\MathSoln

Framework

.NET Framework 4.8

Project will be created in "D:\Nilesh\Feb25\DAC\dotnet\day01\MathSoln\MathClient\"

Back

Create



Search

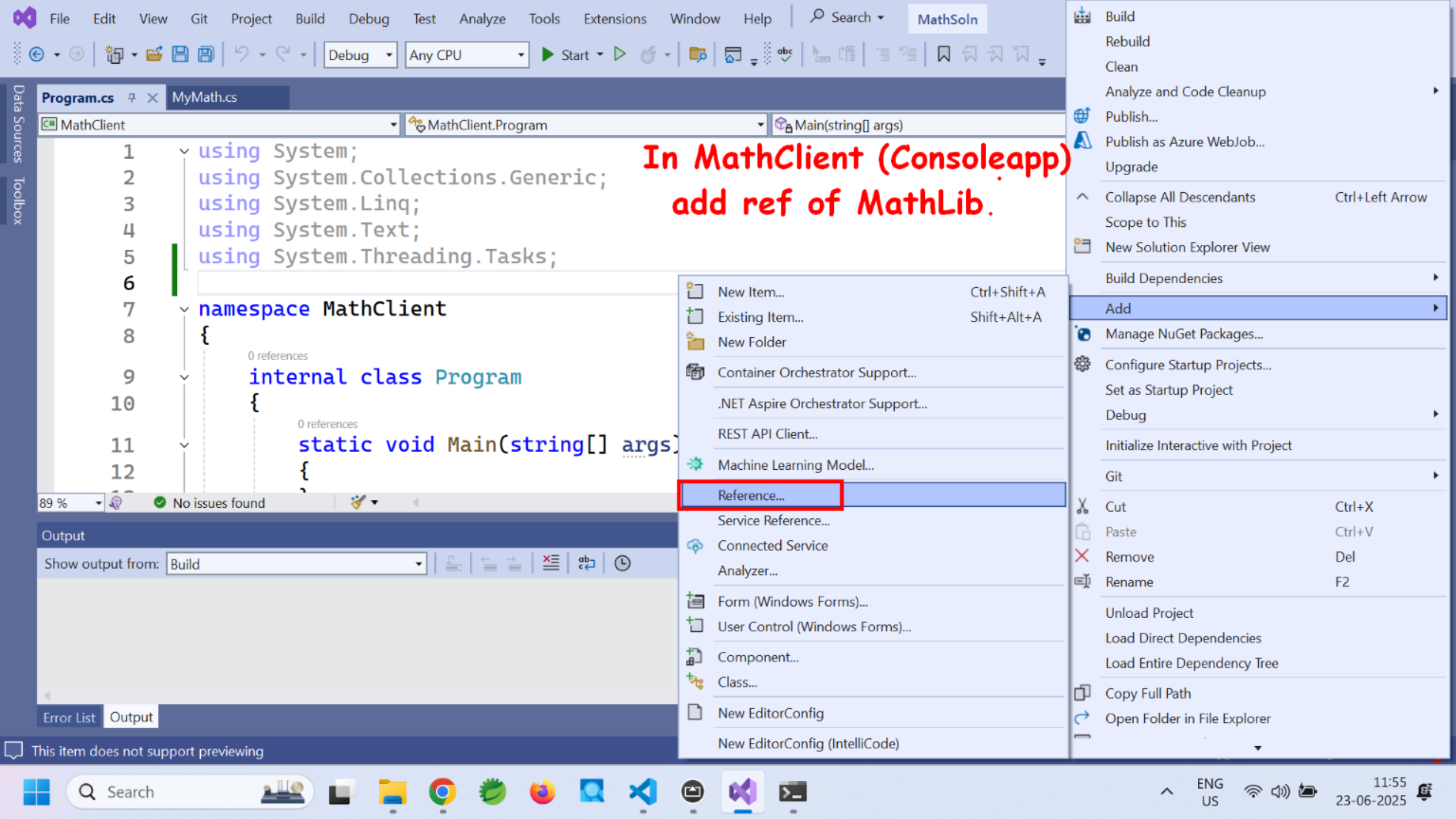


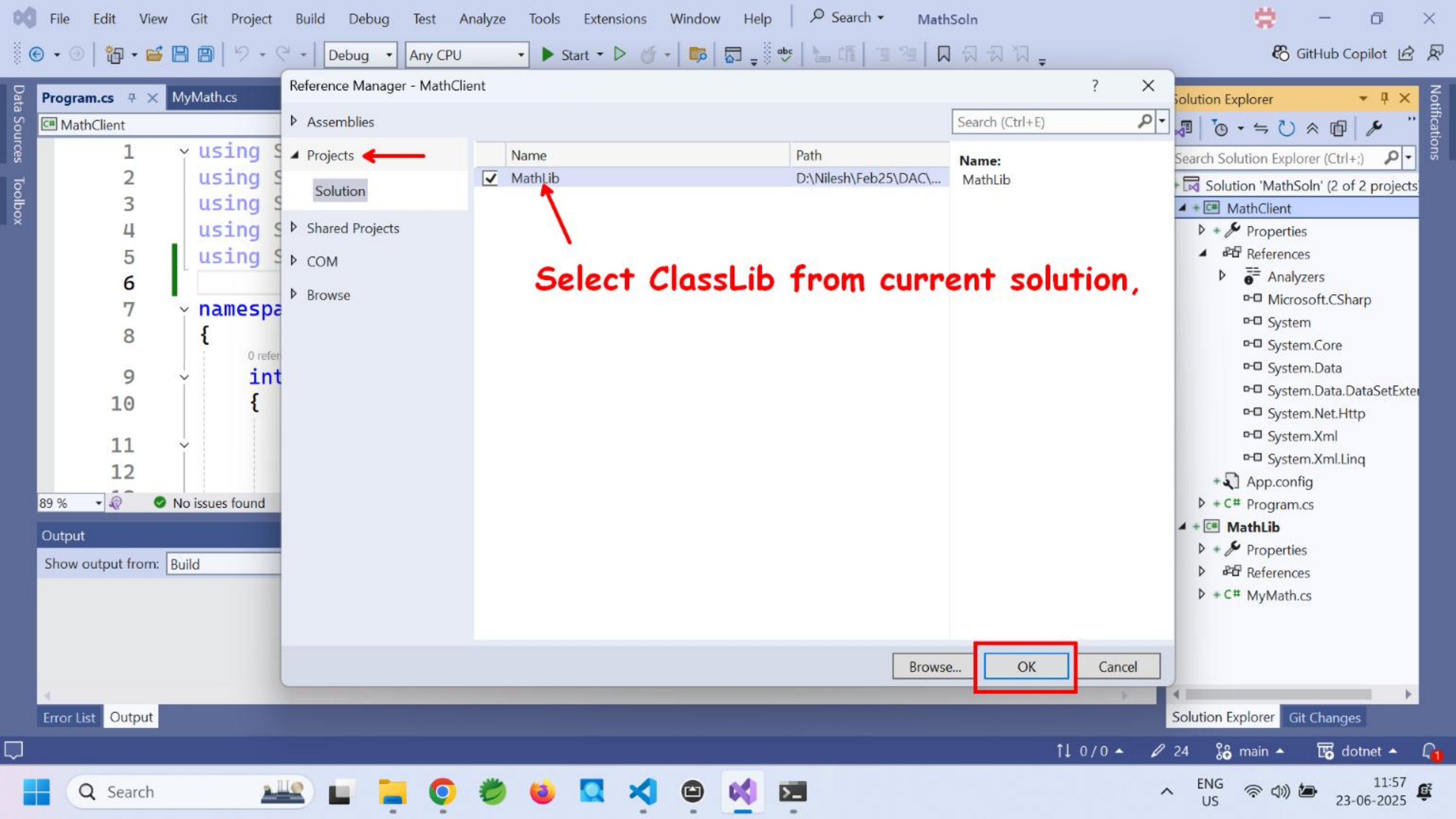
ENG
US

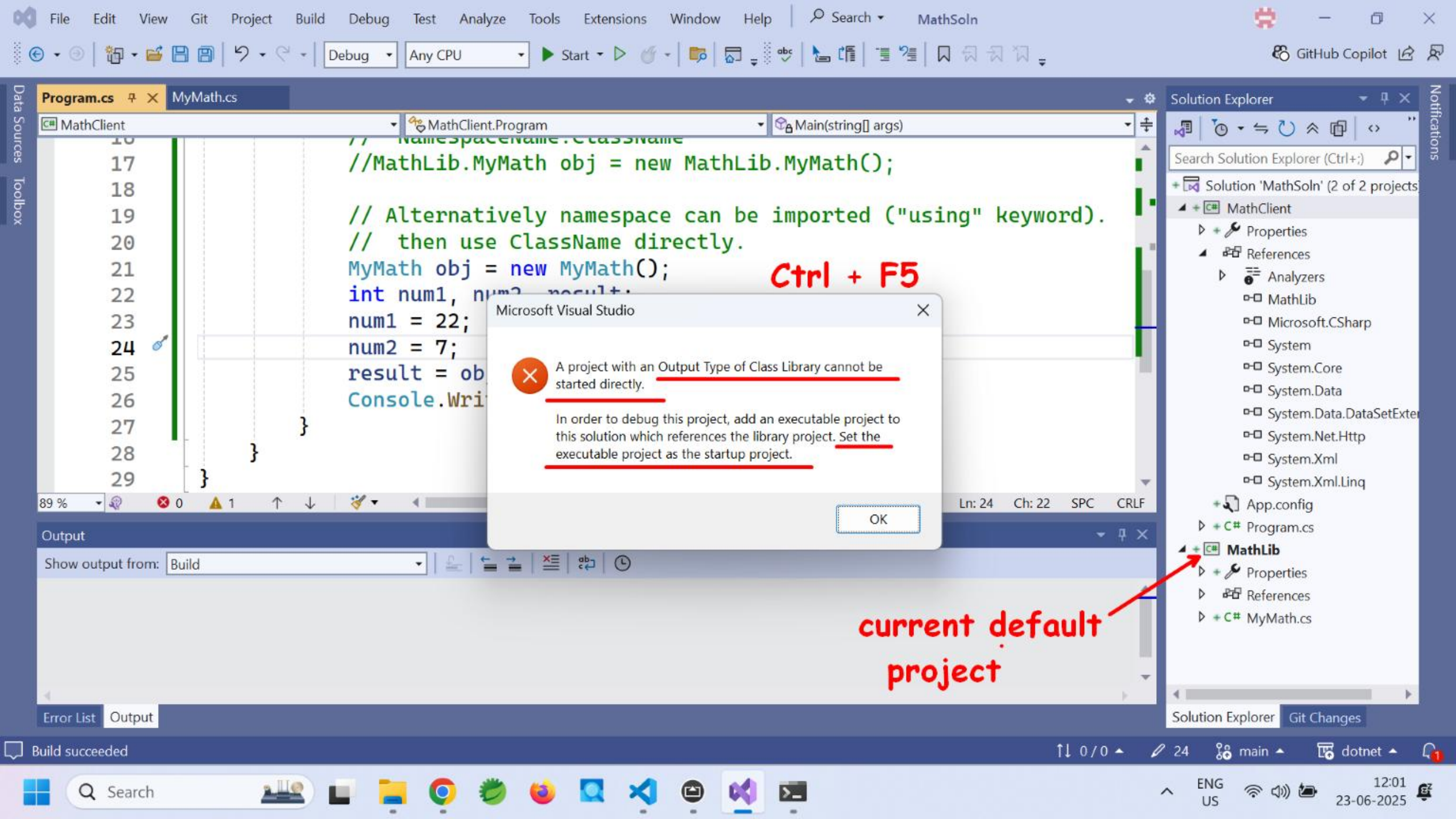


11:53
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File Edit View Git Project Build Debug Test Analyze Tools Extensions Window Help Search MathSoln

Program.cs MyMath.cs

MathClient MathClient.Program Main(string[] args)

```
16 // namespaceName.ClassName
17 //MathLib.MyMath obj = new MathLib.MyMath();
18
19 // Alternatively namespace can be imported ("using" keyword)
20 // then use ClassName directly.
21 MyMath obj = new MyMath();
22 int num1, num2, result;
23 num1 = 22;
24 num2 = 7;
25 result = obj.Multiply(num1, num2);
26 Console.WriteLine("Result: " + result);
27 }
28 }
29 }
```

89 % 0 1 Ln: 24 Ch: 22

Output

Show output from: Build

Error List Output

This item does not support previewing

MathClient proj

Build

Rebuild

Clean

Analyze and Code Cleanup

Publish...

Publish as Azure WebJob...

Upgrade

Collapse All Descendants Ctrl+Left Arrow

Scope to This

New Solution Explorer View

Build Dependencies

Add

Manage NuGet Packages...

Configure Startup Projects...

Set as Startup Project

Debug

Initialize Interactive with Project

Git

Cut Ctrl+X

Paste Ctrl+V

Remove Del

Rename F2

Unload Project

Load Direct Dependencies

Load Entire Dependency Tree

Copy Full Path

Open Folder in File Explorer



Search

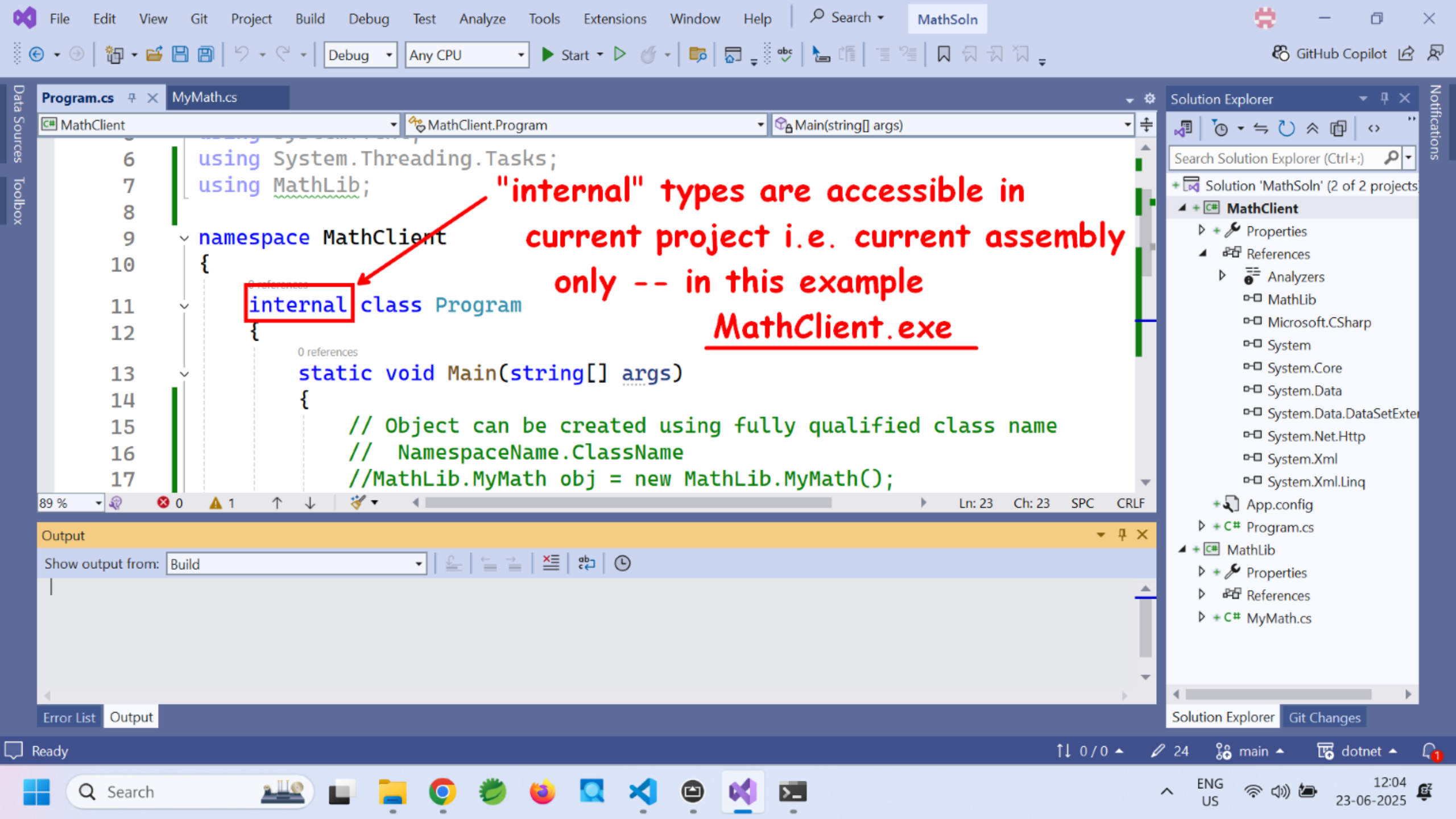


ENG
US



12:02
23-06-2025





"internal" types are accessible in current project i.e. current assembly only -- in this example MathClient.exe

```
Program.cs | MyMath.cs
C# MathClient | MathClient.Program | Main(string[] args)
6 using System.Threading.Tasks;
7 using MathLib;
8
9 namespace MathClient
10 {
11     internal class Program
12     {
13         static void Main(string[] args)
14         {
15             // Object can be created using fully qualified class name
16             // NamespaceName.ClassName
17             //MathLib.MyMath obj = new MathLib.MyMath();
18         }
19     }
20 }
```

Solution Explorer

Search Solution Explorer (Ctrl+;)

- Solution 'MathSoln' (2 of 2 projects)
 - MathClient
 - Properties
 - References
 - Analyzers
 - MathLib
 - Microsoft.CSharp
 - System
 - System.Core
 - System.Data
 - System.Data.DataSetExtensions
 - System.Net.Http
 - System.Xml
 - System.Xml.Linq
 - App.config
 - MathLib
 - Properties
 - References
 - MyMath.cs

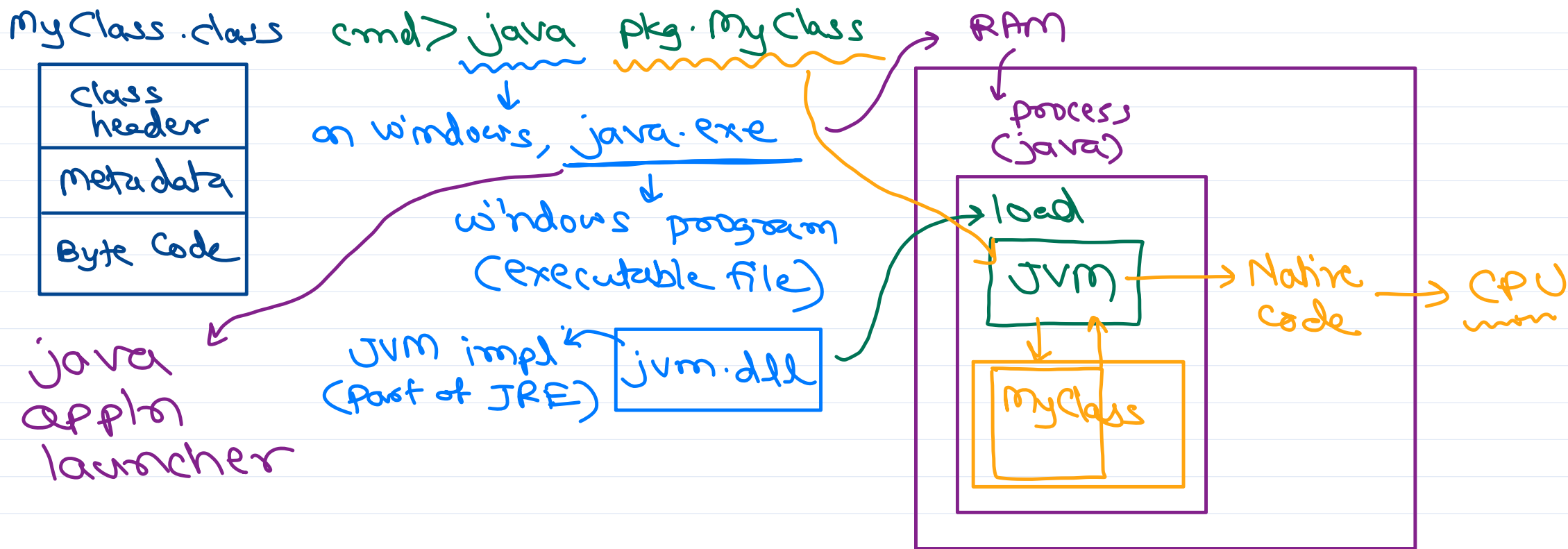
Solution Explorer | Git Changes

Output

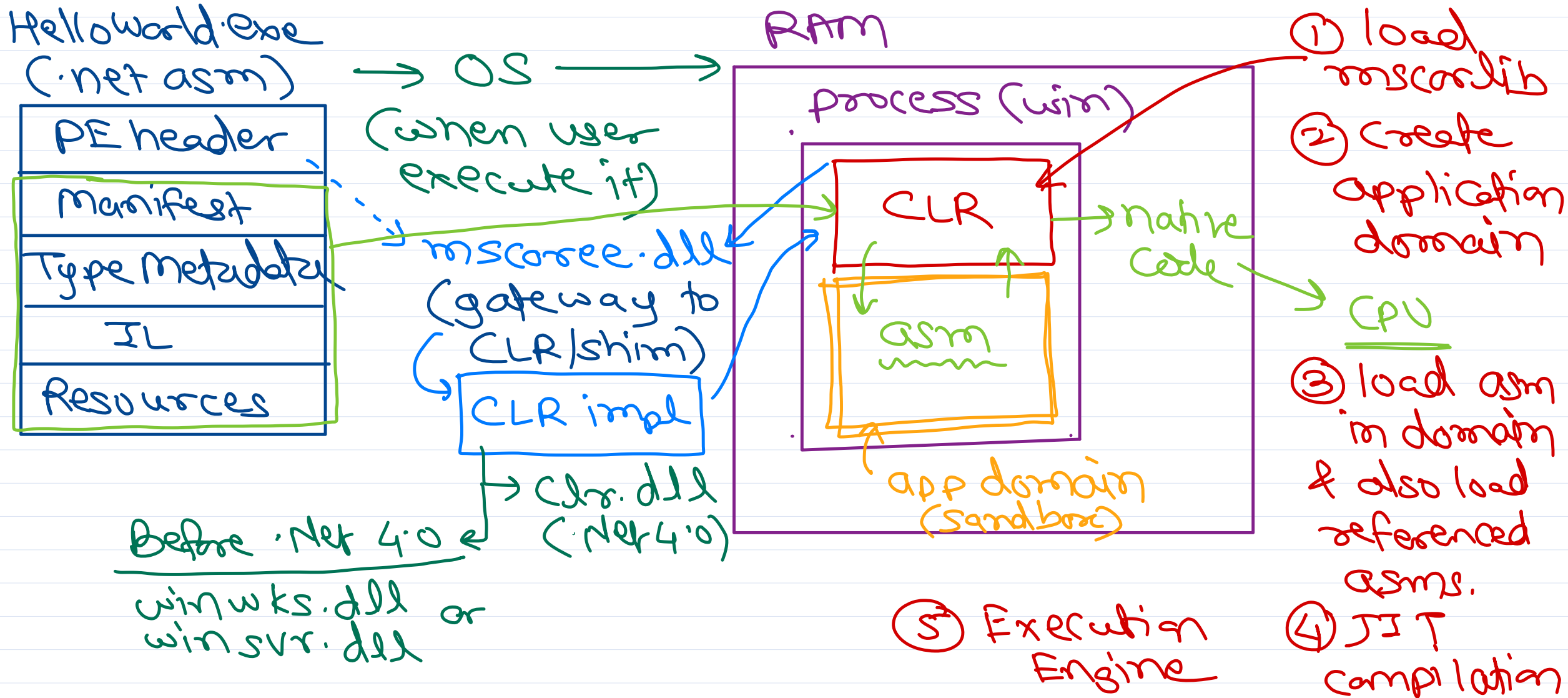
Show output from: Build

Error List | Output

Java execution



.Net program execution



Create a new project

Recent project templates

-  Console App (.NET Framework) C#
-  Class Library (.NET Framework) C#
-  Blank Solution
-  Console App C#
-  Windows Forms App C#

Search for templates (Alt+S)

Clear all

C#

Windows

Console



Console App

A project for creating a command-line application that can run on .NET on Windows, Linux and macOS

C#

Linux

macOS

Windows

Console



Console App (.NET Framework)

A project for creating a command-line application

C#

Windows

Console

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Next



Search

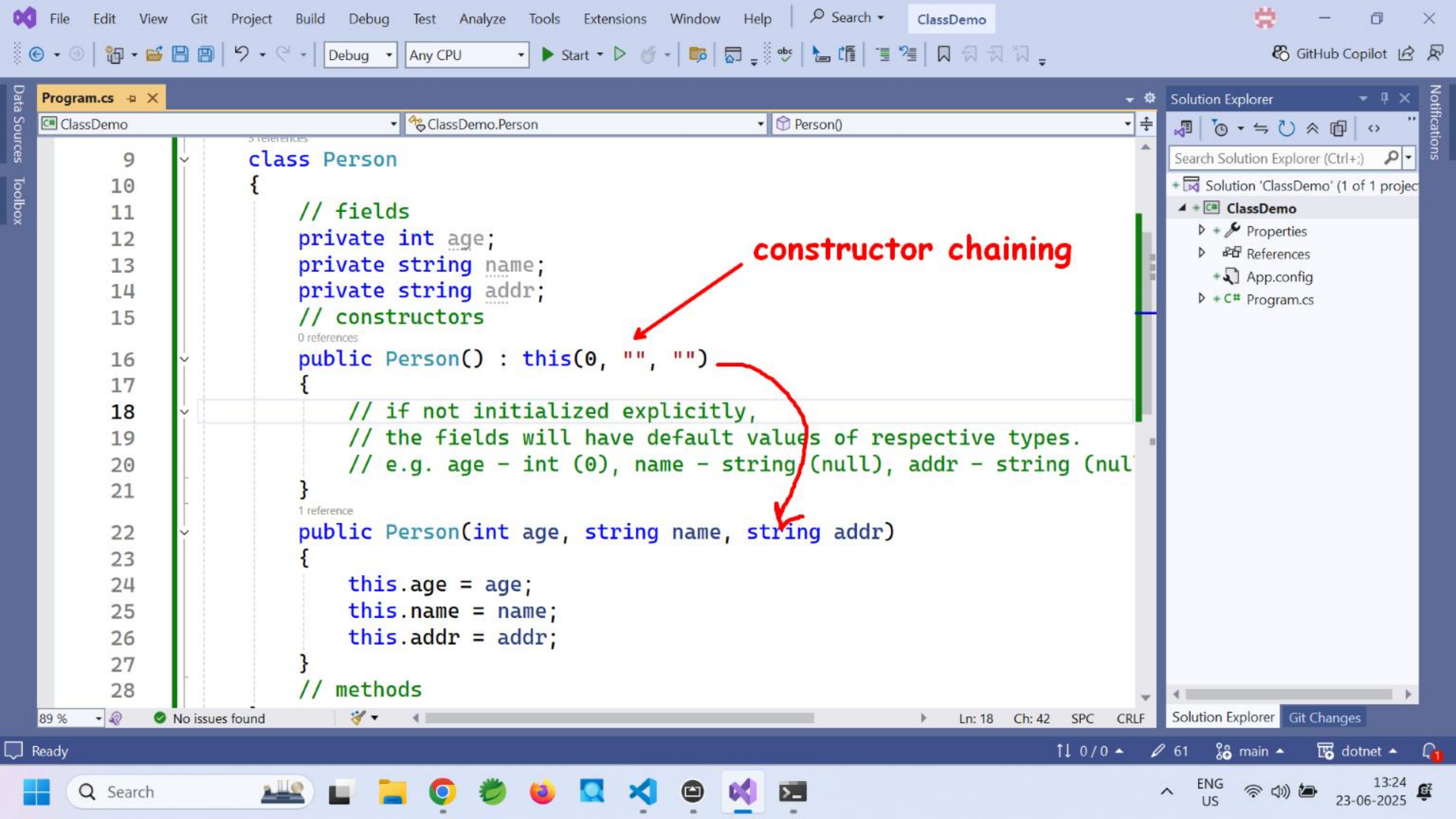


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13:18
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Thank you!

Nilesh Ghule <nilesh@sunbeaminfo.com>

